TP1- implement inter-VLAN Routing

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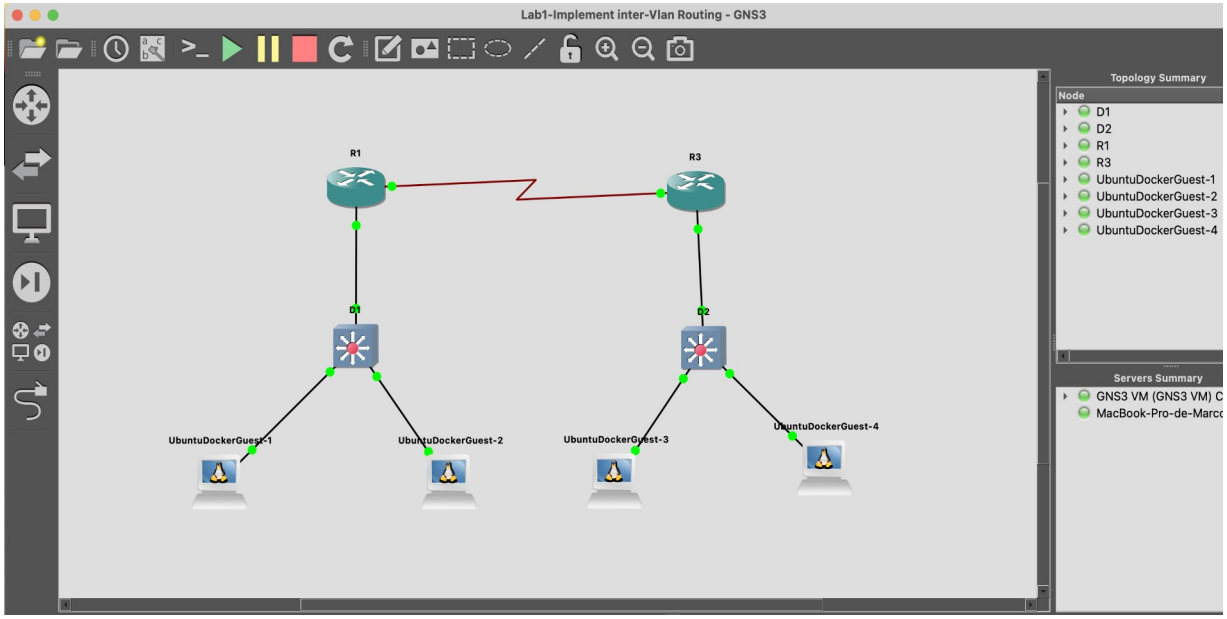
Topology



Config and results

```
Bmarco11 — D1 — telnet 192.168.71.128 5002 — 122×24
D1(config-if-range)#
*Oct 26 09:37:32.221: %LINK-5-CHANGED: Interface Ethernet2/2, changed state to administratively down
*Oct 26 09:37:32.221: %LINK-5-CHANGED: Interface Ethernet2/3, changed state to administratively down
*Oct 26 09:37:32.230: %LINK-5-CHANGED: Interface Ethernet3/0, changed state to administratively down
*Oct 26 09:37:32.230: %LINK-5-CHANGED: Interface Ethernet3/1, changed state to administratively down
*Oct 26 09:37:32.230: %LINK-5-CHANGED: Interface Ethernet3/2, changed state to administratively down
*Oct 26 09:37:32.230: %LINK-5-CHANGED: Interface Ethernet3/3, changed state to administratively down
*Oct 26 09:37:33.217: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to down
*Oct 26 09:37:33.217: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/1, changed state to down
*Oct 26 09:37:33.217: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to down
*Oct 26 09:37:33.217: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/3, changed state to down
*Oct 26 09:37:33.222: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to down
*Oct 26 09:37:33.222: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to down
D1(config-if-range)#
*Oct 26 09:37:33.222: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed state to down
*Oct 26 09:37:33.222: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed state to down
*Oct 26 09:37:33.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/0, changed state to down
*Oct 26 09:37:33.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/1, changed state to down
*Oct 26 09:37:33.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/2, changed state to down
*Oct 26 09:37:33.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/3, changed state to down
D1(config-if-range)#exit
D1(config)#exit
D1#
*Oct 26 09:37:55.381: %SYS-5-CONFIG_I: Configured from console by console
```

Bmarco11 — D2 — telnet 192.168.71.128 5003 — 80×24

```
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#hostname D2
D2(config)#line con 0
D2(config-line)#exec-timeout 0 0
D2(config-line)#logging synchronous
D2(config-line)#exit
D2(config)#interface range e1/0-3, e2/0-3, e3/0-3
D2(config-if-range)#shutdown
D2(config-if-range)#
*Oct 26 09:43:07.052: %LINK-5-CHANGED: Interface Ethernet1/0, changed state to a
dministratively down
*Oct 26 09:43:07.054: %LINK-5-CHANGED: Interface Ethernet1/1, changed state to a
dministratively down
*Oct 26 09:43:07.056: %LINK-5-CHANGED: Interface Ethernet1/2, changed state to a
dministratively down
*Oct 26 09:43:07.060: %LINK-5-CHANGED: Interface Ethernet1/3, changed state to a
dministratively down
*Oct 26 09:43:07.060: %LINK-5-CHANGED: Interface Ethernet2/0, changed state to a
dministratively down
*Oct 26 09:43:07.060: %LINK-5-CHANGED: Interface Ethernet2/1, changed state to a
dministratively down
D2(config-if-range)#
*Oct 26 09:43:07.070: %LINK-5-CHANGED: Interface Ethernet2/2, changed state to a
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 80×24

```
Trying 192.168.71.128...
Connected to 192.168.71.128.
Escape character is '^]'.

IOU1#enable
IOU1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
IOU1(config)#hostname R1
R1(config)#line con 0
R1(config-line)#exec-timeout 0 0
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#
```

Bmarco11 — R3 — telnet 192.168.71.128 5001 — 80×24

```
Trying 192.168.71.128...
Connected to 192.168.71.128.
Escape character is '^]'.

IOU2#
IOU2#enable
IOU2#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
IOU2(config)#hostname R3
R3(config)#line con 0
R3(config-line)#exec-timeout 0 0
R3(config-line)#logging synchronous
R3(config-line)#exit
R3(config)#
```

Bmarco11 — D1 — telnet 192.168.71.128 5002 — 79×24

```
D1(config-if)#no shutdown
D1(config-if)#exit
D1(config)#interface e0/2
D1(config-if)#switchport mode access
D1(config-if)#switchport access vlan 60
D1(config-if)#no shutdown
D1(config-if)#exit
D1(config)#interface vlan 50
D1(config-if)#ip
*Oct 26 09:57:58.994: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, c
hanged state to down
D1(config-if)#ip address 10.2.50.1 255.255.255.0
D1(config-if)#ipv6 address fe80::d1:2 link-local
D1(config-if)#ipv6 address 2001:db8:acad:1050::d1/64
D1(config-if)#no shutdown
D1(config-if)#exit
*Oct 26 09:59:49.630: %LINK-3-UPDOWN: Interface Vlan50, changed state to up
*Oct 26 09:59:50.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, c
hanged state to up
D1(config-if)#exit
D1(config)#interface vlan60
D1(config-if)#
*Oct 26 10:00:03.814: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, c
hanged state to down
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 80×24

```
R1#sho
*Oct 26 12:03:42.931: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip cef
Prefix                Next Hop              Interface
0.0.0.0/0             10.1.3.3              Serial2/0
0.0.0.0/8             drop
0.0.0.0/32            receive
10.1.3.0/24           attached              Serial2/0
10.1.3.0/32           receive               Serial2/0
10.1.3.1/32           receive               Serial2/0
10.1.3.3/32           10.1.3.3              Serial2/0
10.1.3.255/32         receive               Serial2/0
10.1.13.0/24          attached              Ethernet0/0
10.1.13.0/32          receive               Ethernet0/0
10.1.13.1/32          receive               Ethernet0/0
10.1.13.13/32         10.1.13.13            Ethernet0/0
10.1.13.255/32        receive               Ethernet0/0
10.2.0.0/16           10.1.13.13            Ethernet0/0
127.0.0.0/8           drop
224.0.0.0/4           drop
224.0.0.0/24          receive
240.0.0.0/4           drop
255.255.255.255/32    receive
R1#
```

Bmarco11 — root@UbuntuDockerGuest-1: ~ — telnet 192.168.71.128 500...

```
Escape character is '^]'.
UbuntuDockerGuest-1 console is now available... Press RETURN to get started.
root@UbuntuDockerGuest-1:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.50.50  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 2001:db8:acad:1050:42:14ff:fecb:4e00  prefixlen 64  scopeid 0x0<gl
obal>
        inet6 fe80::42:14ff:fecb:4e00  prefixlen 64  scopeid 0x20<link>
        ether 02:42:14:cb:4e:00  txqueuelen 1000  (Ethernet)
        RX packets 7  bytes 478 (478.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7  bytes 602 (602.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@UbuntuDockerGuest-1:~#
```

Bmarco11 — root@UbuntuDockerGuest-2: ~ — telnet 192.168.71.128 500...

```
Escape character is '^]'.
UbuntuDockerGuest-2 console is now available... Press RETURN to get started.
root@UbuntuDockerGuest-2:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.60.50  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 2001:db8:acad:1060:42:feff:fed6:e400  prefixlen 64  scopeid 0x0<gl
obal>
        inet6 fe80::42:feff:fed6:e400  prefixlen 64  scopeid 0x20<link>
        ether 02:42:fe:d6:e4:00  txqueuelen 1000  (Ethernet)
        RX packets 5  bytes 358 (358.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7  bytes 602 (602.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@UbuntuDockerGuest-2:~#
```

Bmarco11 — root@UbuntuDockerGuest-1: ~ — telnet 192.168.71.128 5004 — 86...

```
64 bytes from 10.2.60.50: icmp_seq=52 ttl=63 time=3.34 ms
64 bytes from 10.2.60.50: icmp_seq=53 ttl=63 time=6.73 ms
64 bytes from 10.2.60.50: icmp_seq=54 ttl=63 time=5.13 ms
64 bytes from 10.2.60.50: icmp_seq=55 ttl=63 time=5.01 ms
64 bytes from 10.2.60.50: icmp_seq=56 ttl=63 time=4.92 ms
64 bytes from 10.2.60.50: icmp_seq=57 ttl=63 time=1.93 ms
64 bytes from 10.2.60.50: icmp_seq=58 ttl=63 time=5.27 ms
64 bytes from 10.2.60.50: icmp_seq=59 ttl=63 time=5.33 ms
64 bytes from 10.2.60.50: icmp_seq=60 ttl=63 time=7.71 ms
64 bytes from 10.2.60.50: icmp_seq=61 ttl=63 time=1.33 ms
64 bytes from 10.2.60.50: icmp_seq=62 ttl=63 time=0.474 ms
64 bytes from 10.2.60.50: icmp_seq=63 ttl=63 time=5.10 ms
64 bytes from 10.2.60.50: icmp_seq=64 ttl=63 time=6.71 ms
64 bytes from 10.2.60.50: icmp_seq=65 ttl=63 time=4.84 ms
64 bytes from 10.2.60.50: icmp_seq=66 ttl=63 time=5.22 ms
64 bytes from 10.2.60.50: icmp_seq=67 ttl=63 time=4.98 ms
64 bytes from 10.2.60.50: icmp_seq=68 ttl=63 time=5.09 ms
64 bytes from 10.2.60.50: icmp_seq=69 ttl=63 time=4.96 ms
64 bytes from 10.2.60.50: icmp_seq=70 ttl=63 time=5.09 ms
^C
--- 10.2.60.50 ping statistics ---
70 packets transmitted, 69 received, 1.42857% packet loss, time 69305ms
rtt min/avg/max/mdev = 0.474/4.070/11.876/2.251 ms
root@UbuntuDockerGuest-1:~#
```

```
D1#show mac address-table dynamic
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  50    0242.14cb.4e00    DYNAMIC     Et0/0
  60    0242.fed6.e400    DYNAMIC     Et0/2
Total Mac Addresses for this criterion: 2
D1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 104×24

```
R1(config)#
*Oct 26 10:51:59.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
R1(config)#inerface s2/0
             ^
% Invalid input detected at '^' marker.

R1(config)#interface s2/0
R1(config-if)#ip address 10.1.3.1 255.255.255.0
R1(config-if)#ipv6 address fe80::1:2 link-local
R1(config-if)#ipv6 address 2001:db8:acad:1013::1/64
R1(config-if)#no shutdown
R1(config-if)#exit
*Oct 26 10:54:01.183: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
*Oct 26 10:54:02.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1(config-if)#exit
R1(config)#
*Oct 26 10:54:26.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
R1(config)#ip route 10.2.0.0 255.255.255.0 10.1.13.13
R1(config)#ipv6 route 2001:db8:acad:1050::/64 2001:db8:acad:10d1::d1
R1(config)#ipv6 route 2001:db8:acad:1060::/64 2001:db8:acad:10d1::d1
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.1.3.3
R1(config)#ipv6 route ::/0 2001:db8:acad:1013::3
R1(config)#
```

Bmarco11 — root@UbuntuDockerGuest-3: ~ — telnet 192.168.71.128 500...

```
Trying 192.168.71.128...
Connected to 192.168.71.128.
Escape character is '^]'.
UbuntuDockerGuest-3 console is now available... Press RETURN to get started.
root@UbuntuDockerGuest-3:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.75.50  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 fe80::42:61ff:fe1c:3300  prefixlen 64  scopeid 0x20<link>
        ether 02:42:61:1c:33:00  txqueuelen 1000  (Ethernet)
        RX packets 16  bytes 1156 (1.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 656 (656.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@UbuntuDockerGuest-3:~#
```

Bmarco11 — R3 — telnet 192.168.71.128 5001 — 80×24

```
R3(config)#interface e0/0.85
R3(config-subif)#encapsulation dot1q 85
R3(config-subif)#ip address 10.3.85.1 255.255.255.0
R3(config-subif)#ipv6 address fe80::3:3 link-local
R3(config-subif)#ipv6 address 2001:db8:acad:3085::1/64
R3(config-subif)#no shutdown
R3(config-subif)#exit
R3(config)#interface e0/0.999
R3(config-subif)#encapsulation dot1q 999 native
R3(config-subif)#no shutdown
R3(config-subif)#exit
R3(config)#interface s2/0
R3(config-if)#ip address 10.1.3.3 255.255.255.0
R3(config-if)#ipv6 address fe80::3:1 link-local
R3(config-if)#ipv6 address 2001:db8:acad:1013::3/64
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Oct 26 11:43:46.975: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
*Oct 26 11:43:47.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0,
 changed state to up
R3(config)#ip route 0.0.0.0 0.0.0.0 10.1.3.1
R3(config)#ipv6 route ::/0 2001:db8:acad:1013::1
R3(config)#
```